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Midterm Examination
CHE 230
Molecular Systems in Chemical Engineering

UIC Chemical Engineering Department

03/02/2026

Time: 70 minutes

Honor Code: As a commitment to the personal, academic, and professional standards upheld by the Honors College, as a student, I hereby pledge to respect, preserve, and promote the Honor Code of Conduct during my entire tenure at the Honors College. I agree to uphold high ethical principles in all personal, academic, and professional endeavors related to the Honors College and beyond, and to always be mindful of being a representative of the Honors College and the UIC community

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Date 3/2/2026

Section 1: Fundamental to Molecular Systems (30 points)

17-

1. (5 points) What is the difference between a physical and a chemical^{bond}? Give one example of each.

physical bond $\rightarrow$ breaks / changes due to things like temperature, pressure, density

Chemical bond $\rightarrow$ attraction between electrons (ex: covalent bond). ³

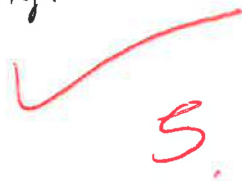


2. (5 points) Graphite and diamond are chemically identical yet have different physical properties. Please briefly explain why.

Graphite and diamond are polymorphs of carbon (same chemical composition but different atomic arrangement).

Diamond: each C atom forms 4 sp^3 covalent bonds in a tetrahedral 3D network; strong, hard, electrical insulator.

Graphite: each C atom forms 3 sp^2 bonds in hexagonal sheets with van der Waals forces; soft, conducts electricity.



3. (5 points) Briefly cite the main differences between ionic, covalent, and metallic bonding.

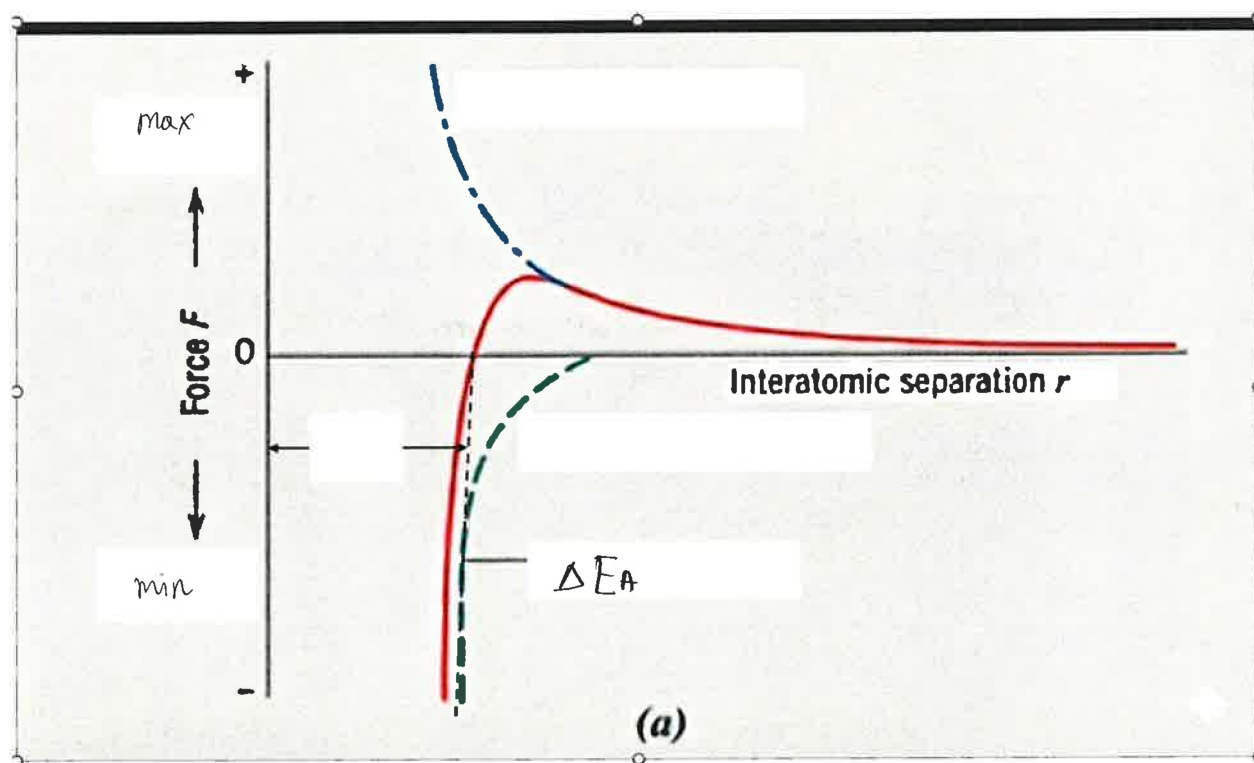
Bond type	Electron behavior
ionic	e^- transferred; $\oplus$ & $\ominus$ ions attract
covalent	e^- shared between atoms
metallic	"sea" of e^- shared among atoms

✓ 5

4. (6 points) The graph provided illustrates the forces acting between two atoms as a function of their interatomic separation, r .

Fill in the blank boxes on the diagram with the correct terms.

X



5. (9 points) The atomic radii of $\overset{Z_1}{\text{Mg}^{2+}}$ and $\overset{Z_2}{\text{O}^{2-}}$ ions are $\overset{r_1}{0.072}$ nm and $\overset{r_2}{0.140}$ nm, respectively.

(a) Calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another)

(b) What is the force of repulsion at this same separation distance?

Given Constants:

- Magnitude of electrical charge for an electron (e): 1.602×10^{-19} C
- Permittivity of a vacuum ϵ_0 : 8.854×10^{-12} F/m (or $\text{C}^2/(\text{N}\cdot\text{m}^2)$)

a) $F_A = \frac{1}{4\pi\epsilon_0 r^2} (|z_1|e)(|z_2|e), r = r_1 + r_2 \quad 1\text{nm} = 10^{-9}\text{m}$

$F_A = 8.854 \times 10^{-12} (2 \cdot e)(2 \cdot e) = 9.08 \cdot 10^{-12} \text{ N}$

b) $F_A = -9.08 \cdot 10^{-12} \text{ N}$

Section 2: Crystal Solid (25 points)

23

1. (5 points) Calculate the radius of an iridium atom, given that Ir has an FCC crystal structure, a density of 22.4 g/cm^3 , and an atomic weight of 192.2 g/mol ($N_A = 6.022 \times 10^{23} \text{ atoms/mol}$).

$$\rho = \frac{n A}{V_c N_A} \rightarrow V_c = \frac{n A}{\rho N_A} = \frac{4 \cdot 192.2 \text{ g/mol}}{22.4 \text{ g/cm}^3 \cdot 6.022 \cdot 10^{23} \text{ atoms/mol}} = 5.70 \cdot 10^{-23} \text{ cm}^3$$

$$V_c = 16\sqrt{2} r^3$$

$$r^3 = \frac{V_c}{16\sqrt{2}} = 2.52 \cdot 10^{-24}$$

$$r = 1.3606 \cdot 10^{-8} \text{ cm}$$

✓5

2. (10 points) List the point coordinates for all atoms that are associated with the FCC unit cell (Figure 2)

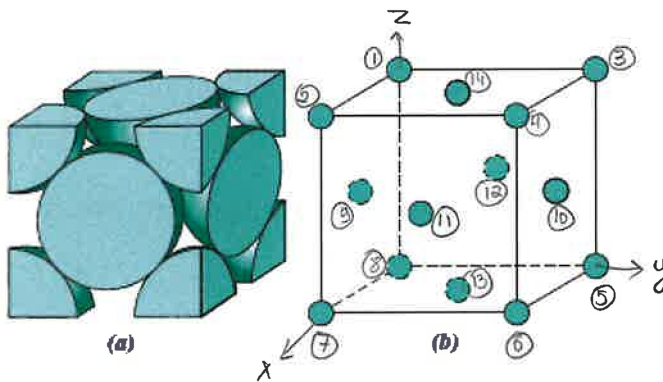


Figure 2 – FCC structure

(x, y, z)

- 1 (0, 0, 1)
- 2 (1, 0, 1)
- 3 (0, 1, 1)
- 4 (1, 1, 1)
- 5 (0, 1, 0)
- 6 (1, 0, 0)
- 7 (1, 0, 0)
- 8 (0, 0, 0)
- 9 ($1/2, 0, 1/2$)
- 10 ($1/2, 1, 1/2$)
- 11 ($1, 1/2, 1/2$)
- 12 ($0, 1/2, 1/2$)
- 13 ($1/2, 1/2, 0$)
- 14 ($1/2, 1/2, 1$)

✓10

3. (5 points) What is the unit cell number, coordination number, Edge Length and APF of a Body Centered Cubic (Show calculation of APF).

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8} + \frac{N_c}{8} = 2$$

$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{V_{atoms}}{a^3} = 0.68$$

4. (5 points) Match each definition or description below with the most appropriate term from the word bank. Each term may be used once, more than once, or not at all.

Word Bank

Grain Boundary	Substitutional impurity	Single Crystal	Vacancy Defect
Interstitial Impurity	Coordination Number	Edge Length	Unit Cell

Coordination number The number of nearest neighbor or touching atom.

Interstitial impurity A two-dimensional planar defect that separates regions of identical crystal structure but differing crystallographic orientations.

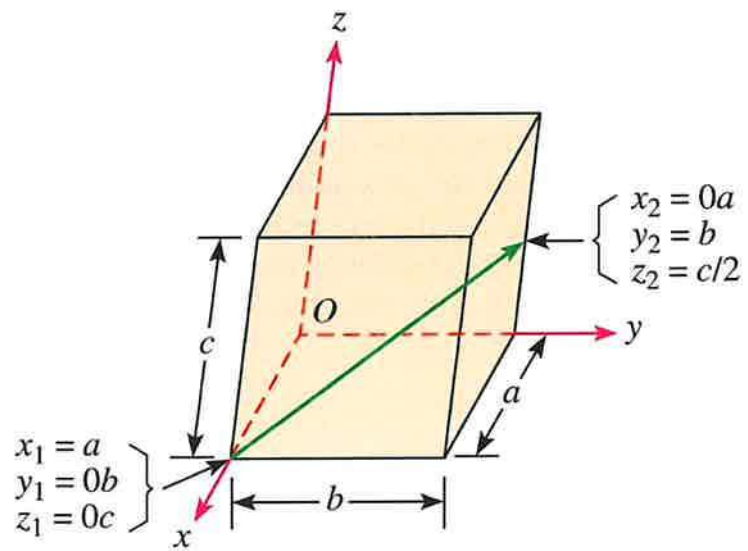
Substitutional impurity Foreign atom replaces host atom.

Edge length The physical dimension or length of one side of a unit cell.

Single Crystal A crystalline solid in which the perfectly ordered, repeating arrangement of atoms is continuous and unbroken throughout the entire sample.

Extra Credit

(5 points) Determine the indices for the direction shown in the accompanying figure.



Section 3. Material Properties (45 points) 39.

Diffusion

1. (10 points) For some applications, it is necessary to harden the surface of a steel (or iron-carbon alloy) above that of its interior. One way this may be accomplished is by increasing the surface concentration of carbon in a process termed carburizing; the steel piece is exposed, at an elevated temperature, to an atmosphere rich in a hydrocarbon gas, such as methane (CH_4). Consider one such alloy that initially has a uniform carbon concentration of 0.25 wt% and is to be treated at 950°C (1750°F). If the concentration of carbon at the surface is suddenly brought to and maintained at 1.20 wt%, how long will it take to achieve a carbon content of 0.80 wt% at a position 0.5 mm below the surface? The diffusion coefficient for carbon in iron at this temperature is $1.6 \times 10^{-11} \text{ m}^2/\text{s}$; assume that the steel piece is semi-infinite.

D

Error Function is on the last page

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\frac{0.80 - 0.25}{1.20 - 0.25} = 1 - \text{erf}(z)$$

$$0.421053 = \text{erf}(z)$$

$$z \approx 0.39$$

$$0.39 = \frac{x}{2\sqrt{Dt}} \quad x = 0.5 \text{ mm} = 0.5 \cdot 10^{-3} \text{ m}$$

$$2\sqrt{Dt} = \frac{x}{0.39} = 0.001282$$

$$\sqrt{Dt} = 0.000641$$

$$Dt = 4.10914 \cdot 10^{-7}$$

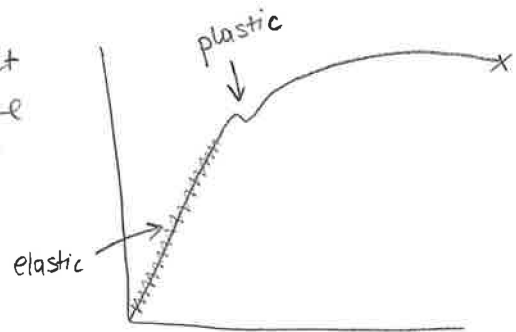
$$t = 25682.1 \text{ s}$$

✓ 10

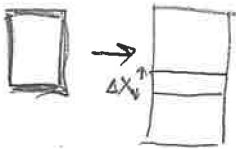
2. (4 points) **Distinguish** between elastic and plastic deformations, both by definition, and in terms of behavior on a stress-strain plot. **Draw** a sketch to illustrate your answer.

Elastic deformation $\rightarrow$ reversible, imagine a metal just stretching without necking or any other damage

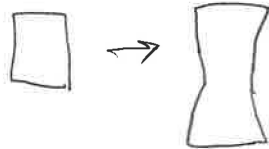
Plastic deformation occurs by movement of dislocations along slip planes



Elastic



Plastic



✓ 4

3. (3 points) **Highlight** the differences between true stress and engineering stress.

$$\sigma = \frac{F}{A}$$

engineering stress

$$\sigma = \frac{F}{A_{\text{instantaneous}}}$$

true stress

✓ 3

4. (8 points) Of the metals listed in the Table below,

- (a) Which will experience the greatest percent reduction in area? Why?
- (b) Which is the strongest? Why?
- (c) Which is the stiffest? Why?
- (d) Which is the hardest? Why?

Material	Yield Strength (MPa)	Tensile Strength (MPa)	Strain at Fracture	Fracture Strength (MPa)	Elastic Modulus (GPa)
A	310	340	0.23	265	210
B	100	120	0.40	105	150
C	415	550	0.15	500	310
D	700	850	0.14	720	210
E	Fractures before yielding			650	350

- (a) B b/c it has largest strain at fracture
- (b) D b/c it has highest tensile strength
- (c) E b/c it has largest elastic modulus
- (d) D b/c it has highest yield strength



5. (10 points) Consider a cylindrical titanium wire 3.0 mm (0.12 in.) in diameter and 2.5×10^4 mm (1000 in.) long. When a load of 500 N (112 lbf) is applied, the wire experiences an elongation of 16.5 mm. Assuming that the deformation is totally elastic, calculate the Young's Modulus of the titanium in GPa.

$$\Delta L = \frac{FL}{AE} \quad E_{Ti} = 107 \cdot 10^3 \text{ Pa} \quad A = \frac{\pi d^2}{4} = \frac{\pi (3.0 \cdot 10^{-3} \text{ m})^2}{4} = 0.00007 \text{ m}^2$$

$$16.5 \cdot 10^{-3} \text{ m} = \frac{500 \text{ N} \cdot 25 \text{ m}}{0.00007 \text{ m}^2 \cdot E}$$

$$E = \frac{FL}{A\Delta L} = 1.07 \cdot 10^{11} \text{ Pa} \\ = 107175 \text{ GPa}$$

$$\text{Pa} = 10^{-6} \text{ GPa}$$

✓ 10

- b. A single crystal of some metal has a τ_{crss} of 30.7 MPa and is exposed to a tensile stress of 61 MPa.

(2 points) (i) Will yielding occur when $\phi = 56^\circ$ and $\lambda = 32^\circ$.

(2 points) (ii) If not, what stress is necessary?

$$(i) \tau = \sigma \cos \phi \cos \lambda = 61 \cos(56^\circ) \cos(32^\circ) = 28.9 \text{ MPa}$$

$28.9 \text{ MPa} < 30.7 \text{ MPa}$, yielding will not occur

$$(ii) \sigma = \frac{\tau_{\text{crss}}}{\cos \phi \cos \lambda} = 64.8 \text{ MPa}$$

✓ 4

6. (6 points) A relatively large plate of a glass is subjected to a tensile stress of 40 MPa. If the specific surface energy and modulus of elasticity for this glass are 0.3 J/m^2 and 69 GPa respectively, determine the maximum length of a surface flaw that is possible without fracture.

$$\sigma = 40 \text{ MPa}$$

$$E = 69 \text{ GPa}$$

$$\sigma = \frac{F}{A}$$

$$\sigma_e (1 + \epsilon_e) = \sigma_T$$

$$\epsilon = \frac{\Delta L}{L}$$



Appendix G: Tables of Error Function

X	ERF	X	ERF	X	ERF	X	ERF
.01	.011283415556	.46	.484655390002	.91	.801882825766	1.36	.945561436561
.02	.022564574692	.47	.493745050886	.92	.806767721548	1.37	.947312398035
.03	.033841222342	.48	.502749670695	.93	.811563538585	1.38	.949016035256
.04	.045111106145	.49	.511668261189	.94	.816271018976	1.39	.950673295805
.05	.056371977797	.50	.520499877813	.95	.820890807273	1.40	.952285119763
.06	.067621594393	.51	.529243619841	.96	.825423649644	1.41	.953852439360
.07	.078857719771	.52	.537898630479	.97	.829870293036	1.42	.955376178641
.08	.090078125841	.53	.546464096935	.98	.834231504340	1.43	.956857253145
.09	.101280593915	.54	.554939250456	.99	.838508069555	1.44	.958296569601
.10	.112462916018	.55	.563323366325	1.00	.842700792950	1.45	.959695025637
.11	.123622896199	.56	.571615763824	1.01	.846810496228	1.46	.961053509513
.12	.134758351820	.57	.579815806164	1.02	.850838017701	1.47	.962372899854
.13	.145867114836	.58	.587922900382	1.03	.854784211454	1.48	.963654065414
.14	.156947033063	.59	.595936497198	1.04	.858649946527	1.49	.964897864843
.15	.167995971427	.60	.603856090848	1.05	.862436106090	1.50	.966105146475
.16	.17901181319	.61	.611681218876	1.06	.866143586635	1.51	.967276748129
.17	.189992461202	.62	.619411461899	1.07	.869773297164	1.52	.968413496920
.18	.200935839019	.63	.627046443338	1.08	.873326158388	1.53	.969516209093
.19	.211839892158	.64	.634585829122	1.09	.876803101938	1.54	.970585689861
.20	.222702589210	.65	.642029327356	1.10	.880205069574	1.55	.971622733262
.21	.233521922982	.66	.649376687963	1.11	.883533012415	1.56	.972628122027
.22	.244295911599	.67	.656627702300	1.12	.886787890165	1.57	.973602627462
.23	.255022599592	.68	.663782202741	1.13	.889970670363	1.58	.974547009343
.24	.265700058954	.69	.670840062235	1.14	.893082327630	1.59	.975462015822
.25	.276326390168	.70	.677801193837	1.15	.896123842937	1.60	.976348383345
.26	.286899723216	.71	.684665550217	1.16	.899096202880	1.61	.977206836583
.27	.297418218547	.72	.691433123139	1.17	.902000398966	1.62	.978038088373
.28	.307880068029	.73	.698103942917	1.18	.904837426915	1.63	.978842839674
.29	.318283495861	.74	.704678077855	1.19	.907608285972	1.64	.979621779524
.30	.328626759459	.75	.711155633654	1.20	.910313978230	1.65	.980375585023
.31	.338908150311	.76	.717536752806	1.21	.912955507973	1.66	.981104921311
.32	.349125994796	.77	.723821613965	1.22	.915533881027	1.67	.981810441565
.33	.359278654974	.78	.730010431299	1.23	.918050104127	1.68	.982492787002
.34	.369364529345	.79	.736103453821	1.24	.920505184299	1.69	.983152586895
.35	.379382053562	.80	.742100964708	1.25	.922900128256	1.70	.983790458591
.36	.389329701129	.81	.748003280598	1.26	.925235941810	1.71	.984407007545
.37	.399205984043	.82	.753810750875	1.27	.927513629295	1.72	.985002827359
.38	.409009453420	.83	.759523756938	1.28	.929734193014	1.73	.985578499828
.39	.418738700070	.84	.765142711455	1.29	.931898632689	1.74	.986134594997
.40	.428392355047	.85	.770668057608	1.30	.934007944941	1.75	.986671671219
.41	.437969090155	.86	.776100268324	1.31	.936063122773	1.76	.987190275231
.42	.447467618426	.87	.781439845491	1.32	.938065155079	1.77	.987690942224
.43	.456886694550	.88	.786687319174	1.33	.940015026158	1.78	.988174195930
.44	.466225115278	.89	.791843246814	1.34	.941913715258	1.79	.988640548706
.45	.475481719787	.90	.796908212423	1.35	.943762196123	1.80	.989090501636

